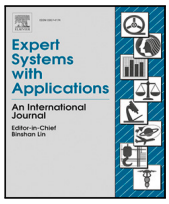
Expert Systems With Applications 280 (2025) 127340



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa





EFJSP-IBDR: Energy-efficient flexible job shop scheduling under incentive-based demand response via graph reinforcement learning with dual attention mechanism

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ARTICLE INFO

Keywords:
Flexible job shop scheduling
Energy-efficient manufacturing
Incentive-based demand response
Graph reinforcement learning
Dual attention mechanism

ABSTRACT

As energy demand grows and the penetration of intermittent renewable energy, optimizing energy consumption patterns in industrial production processes has become an increasing challenge for sustainable manufacturing. This study investigates an energy-efficient flexible job shop scheduling problem with incentive-based demand response (EFJSP-IBDR), aiming to enhance energy consumption efficiency and cost-effectiveness. Given the time-sensitive and complex state features of EFJSP-IBDR, we propose an end-to-end graph reinforcement learning (GRL) approach. Initially, we decompose EFJSP-IBDR into a set of preference-based subproblems to facilitate multi-objective optimization. On this basis, a Markov decision process (MDP) with heterogeneous graph states is employed to model these subproblems. Subsequently, we construct graph neural network schedulers that integrate dual attention mechanisms to extract the features necessary for decision-making in each subproblem. Additionally, a similarity-based parameter transfer strategy is designed to accelerate the training process of all schedulers. Empirical results demonstrate that our approach can reduce peak demand by 47.22% and average load by 51.01% during the manufacturing cycle. The effectiveness of the proposed method suggests its potential to improve load-side energy consumption flexibility and grid resilience.

1. Introduction

Influenced by global economic recovery, geopolitical shocks, and the ongoing energy crisis, there has been considerable volatility in global energy market prices recently. From 2020 to early 2022, global oil prices tripled, and natural gas prices in Europe and Asia surged sixfold (REN21, 2022). In 2023, the average wholesale electricity price in Europe reached approximately 105 euros per megawatt-hour, nearly double the level of 2019 (IEA, 2024b). Concurrently, a significant increase in energy-related CO_2 emissions has intensified the pace of global climate change. In 2023, there was a 1.1% increase in global CO_2 emissions from energy sources, reaching a record high of 37.4 billion tonnes (Gt) (IEA, 2024a). Consequently, end-users are facing significant economic and environmental pressures concerning their energy consumption and associated carbon footprint.

The industrial sector is a major contributor to the growth of final energy consumption (IEA, 2023), possessing a high potential for energy conservation and bearing substantial social responsibility for reducing energy use. Energy-efficient scheduling is a 'soft' technology that significantly improves energy efficiency with less time and capital investment (Li & Wang, 2022), and has been favored by industry and academia in recent years. Moreover, flexible job shops are emerging as one of the most common forms of workshop production organization under Industry 4.0 (Dauzère-Pérès et al., 2023). As cyber–physical production systems (CPPS) and the industrial Internet of Things (IoT) develop, this organizational form increasingly meets the information, communication, and metering infrastructure requirements essential for energy optimization. Therefore, this study focuses on the energy-efficient flexible job shop scheduling problem (EFJSP).

Electricity, as one of the cleanest secondary energy sources, is highly valued for its efficiency and convenience. With the penetration of renewable energy and the "energy prosumer" model emerging (Pham

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https://doi.org/10.1016/j.eswa.2025.127340
Received 18 September 2024; Received in revised form 18 February 2025; Accepted 18 March 2025
Available online 5 April 2025
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et al., 2019), utility operators encounter significant challenges in maintaining the supply–demand balance. Through load-balancing measures such as peak shaving, demand response (DR) has proven effective in mitigating timing mismatches arising from new-generation resources. DR programs are typically categorized into two types: Price-based (PBDR) and Incentive-based (IBDR). PBDR encourages users to adjust their consumption in response to time-varying electricity prices, whereas IBDR offers financial incentives for reducing usage during peak times or grid instability. PBDR such as Time-of-Use (TOU) rates and Critical Peak Pricing (CPP), are non-dispatchable programs designed to reduce predictable, structured peak demands. The timing and extent of utilizing lower electricity prices are entirely consumer-dependent (Otashu & Baldea, 2020). In contrast, IBDR incorporates dynamic elements of response time, power, and price, resulting in more flexible data that better reflects the grid operator's real-time load-shedding requirements. Consequently, IBDR is dispatchable, providing grid operators with greater flexibility in acquiring DR resources (Lu & Hong, 2019). Notably, statistics show that 93% of peak load reductions in the US are due to IBDR (Cappers et al., 2010). Thus, this paper incorporates the IBDR policy in the energy optimization process.

Considering the time sensitivity of IBDR, traditional integer programming methods tend to introduce numerous time slot variables and constraints (Wu et al., 2023). As the complexity of the problem increases, this approach not only affects the compactness and solvability of the scheduling model but also significantly escalates development and maintenance costs. Recently, deep reinforcement learning (DRL) has shown promising results in scheduling problems with complex state features due to its model-free nature and independence from prior domain knowledge (Guo et al., 2024; Lu et al., 2020). Moreover, graph neural networks (GNN) are increasingly applied to various scheduling problems because of their ability to extract decision-related feature embeddings from high-dimensional state spaces and their size-invariant properties (Song et al., 2022). Consequently, the graph reinforcement learning (GRL) approach, which integrates GNN and DRL, emerges as a viable solution for energy-efficient scheduling problems considering IBDR (EFJSP-IBDR).

Considering the issue above, this research introduces a dual attention mechanism-based GRL approach that focuses on tri-objective optimization of makespan, total energy consumption (TEC), and total payload (TPL). Initially, we employ a decomposition strategy to divide the EFJSP-IBDR into several subproblems, each formulated as a Markov Decision Process (MDP). On this basis, the scheduling state is depicted using a heterogeneous disjunctive graph (HDG) that incorporates IBDR features, and the reward is constructed through a bilayer multi-objective aggregated method. Subsequently, we develop a novel graph neural network scheduler that incorporates metapath-edge and heterogeneous attention mechanisms. Through a parameter transfer strategy combined with the Proximal Policy Optimization (PPO) algorithm (Schulman et al., 2017), we collaboratively optimize the model parameters of all schedulers to solve each subproblem. The main contributions of this work are outlined below:

(1) The EFJSP-IBDR is modeled as a set of MDP models with different preferences, and a new heterogeneous graph structure is used to represent the time-varying scheduling states related to production and energy.
(2) A dual attention mechanism-based feature extraction method is designed to reveal the mechanisms of message passing and information fusion within the heterogeneous graph, and the optimal scheduling strategy is learned from the feature embeddings using a similarity-based parameter transfer training algorithm.
(3) A comparison of three different IBDR participation scenarios in real-world cases shows that the proposed method effectively reduces overall energy consumption while securing additional incentive rewards.
(4) Benchmark experiments of various scales indicate that the proposed method outperforms some multi-objective optimization algorithms in terms of solution quality and runtime.

The remainder of this paper is structured as follows. Section 2 offers a comprehensive review of relevant literature. Section 3 introduces the foundational problem model and describes the IBDR mechanism under investigation. Section 4 details the GRL approach for solving EFJSP-IBDR. Section 5 provides the numerical results for a practical case study and benchmark examples. Finally, Section 6 concludes the paper and discusses future research directions.

2. Related work

There are three areas of research closely aligned with our article: energy-efficient scheduling in manufacturing systems, demand response in industrial settings and DRL-based methods in solving FJSP.

Concerning the first, readers can refer to comprehensive literature reviews in Akbar and Irohara (2018), Gahm et al. (2016), Gao et al. (2020). Overall, common methods for energy-saving scheduling include machine speed adjustment (Wei et al., 2022), machine on/off switching (Zhang, Tang, et al., 2017), job-machine allocation (Liu et al., 2017), and batch processing (Liu et al., 2016). Existing studies have validated the feasibility of these strategies and provided extensive management insights. However, optimizing energy consumption patterns rather than levels throughout production operations more effectively meets the practical needs of manufacturers to lower energy costs and grid operators to maintain stable supply margins. Therefore, PBDR and IBDR are increasingly incorporated into research on energy-efficient scheduling.

2.1. Energy-efficient scheduling under PBDR

Energy-efficient scheduling under PBDR seeks to adapt scheduling schemes in response to fluctuating tariffs, with power consumption patterns for industrial end-users largely determined by production decisions. TOU pricing, which is prevalent globally, has been extensively studied in contexts such as single machines (Wu et al., 2023), unrelated parallel machines (Saberi-Aliabad et al., 2020), flow shops (Cui & Lu, 2021), and flexible job shops (Jia et al., 2024) to reduce energy costs. Few studies have employed Real-Time Pricing (RTP), which offers a finer temporal granularity than TOU, thus more accurately reflecting the operational state of the grid. Yun et al. (2023) utilized the flexibility of machine loads under RTP to effectively reduce electricity costs while ensuring production efficiency. To further mitigate the impact of future electricity price uncertainties on production operations, Lu et al. (2021) focused on integrating electricity price forecasts with scheduling decisions to develop a data-driven mechanism for integrated production and energy optimization. Additionally, as renewable technologies advance and costs decrease, traditional manufacturers are gradually transitioning from energy consumers to prosumers, adding extra constraints to the issues above. Ghorbanzadeh and Ranjbar (2023), Liao and Gicquel (2024) examined the energy-saving scheduling that incorporates renewable energy constraints and TOU in various settings, while (Fazli Khalaf & Wang, 2018) investigated scheduling that combines RTP with renewable energy.

These studies indicate that appropriately adjusting production schemes according to electricity pricing policies is an effective way to reduce operational costs for manufacturing enterprises. However, PBDR is a user-centric passive response mode (Füchtenhans & Glock, 2024), making it difficult to predict its impact on power supply–demand balance precisely, and may even lead to adverse effects such as "rebound peaks" (Otashu & Baldea, 2020).

2.2. Energy-efficient scheduling under IBDR

IBDR is a grid-centric active response mode (Füchtenhans & Glock, 2024), and energy-efficient scheduling under IBDR aims to adjust scheduling schemes according to grid incentive signals. Compared to PBDR, there is relatively less research on production scheduling for industrial consumers under IBDR. Studies in this field can be categorized into three types depending on the participation format. Regarding *Ancillary services*, Otashu and Baldea (2020) developed a scheduling approach for chlor-alkali process productivity to offer frequency regulation services. Zhang, Hug, et al. (2017) designed a framework to coordinate cement production loads and onsite storage systems to provide frequency regulation or following services, while (Chau et al., 2017) additionally took solar resources into account. Under *Demand bidding*, Li and Hong (2016) introduced a real-time bidding algorithm for discrete manufacturing that identifies the best bidding strategy by managing machine on/off states. Mohagheghi and Raji (2014) examined a demand bidding model that assists factory managers in determining scheduling schemes while providing a certain available DR capacity. Concerning *Interruptible and curtailable load*, Zhang et al. (2022) proposed a production decision method for random peak periods in the electricity market, focusing on pricing and incentive strategies to reduce operating costs in two-stage production systems. Similarly, Perraudat et al. (2022) optimized total costs under scenarios of random DR reduction requests and varying batch sizes using stochastic planning models. Desta et al. (2018) constructed a management framework for industrial asynchronous production lines to balance productivity and power consumption effectively during DR events. Recently, Weitzel and Glock (2019) built a mixed-integer programming model that generated load reduction curves, quantifying the load reduction potential of a flexible flow shop with a photovoltaic storage system. Füchtenhans and Glock (2024) employed a genetic algorithm to generate the Pareto front, analyzing trade-offs between energy and production objectives in job shops with variable machine speeds across various response modes.

2.3. Solving FJSP via DRL-based methods

In recent years, as computational capabilities has continually increased, DRL approaches have been increasingly adopted to solve FJSP. Following (Lei et al., 2024), we classify recent studies into two main categories: (1) integrating dispatching rules into DRL methods and (2) employing end-to-end learning approaches. In the first category, Du et al. (2022) introduced a Deep Q-Network (DQN) that selects the most appropriate rule from a pool of seven designed dispatching rules to address FJSP with crane transport and setup times, aiming to optimize makespan and TEC. Li et al. (2022) developed a hybrid DQN-based approach to select composite dispatching rules generated via genetic programming when new jobs arrivals and machine failures. Similarly, Wang et al. (2022) utilized two DQNs combined with a real-time processing framework to select composite dispatching rules for each dynamic event, followed by an enhanced local search algorithm to refine the schedule. In contrast, end-to-end DRL methods directly produce scheduling solutions from extracted instance features, without the need to design, combine, or refine dispatching rules. Lei et al. (2022) proposed a multi-action decision framework that integrates GNN and DRL to train action policies for both job operations and machines, yielding high-quality scheduling solutions. Song et al. (2022) improved the disjunctive graph representation of FJSP and introduced a heterogeneous GNN to learn feature embeddings of operations and machines. In another study, Lei et al. (2024) provided an end-to-end hierarchical reinforcement learning framework capable of solving large-scale dynamic FJSP in near real-time. This framework employs a high-level layer for subproblem slicing and two low-level layers dedicated to job operation sequencing and machine assignment. Moreover, Su et al. (2024) suggested a parallel preference-conditioned GRL approach that decomposes a multi-objective FJSP into multiple preference-based subproblems. Notably, a simple multilayer perceptron (MLP) was utilized to facilitate operation and machine message propagation.

In summary, few studies have applied end-to-end learning methodologies to address EFJSP. Furthermore, while most existing work focuses on single-objective formulations or multi-objective transformations through weighting methods, there remains a significant gap in the literature regarding multi-objective DRL approaches that directly produce Pareto sets.

2.4. Research gaps

From the above literature review we can summarize as follows:

(i) From the perspective of IBDR policy, prevailing research predominantly addresses continuous industries and assembly lines, which typically feature stable product processes and consistent machining capabilities. These studies highlight IBDR's potential impact on production operations, focusing primarily on controlling productivity and inventory/buffer utilization at the planning level. However, exploring the demand-side flexibility of flexible job shops and generate the machine-level response schemes remains to be explored.

(ii) From the energy-efficient scheduling problem viewpoint, existing studies typically start with grid operations, translating IBDR into fixed (Desta et al., 2018; Füchtenhans & Glock, 2024) or semi-fixed (Weitzel & Glock, 2019) maximum power consumption limits during scheduling. However, minor adjustments to scheduling schemes in flexible job shops can cause significant shifts in conflicting scheduling objectives. Such changes are often challenging to characterize and address with fixed or manually adjusted linear constraints. Therefore, further incorporating IBDR into optimization objectives and exploring the trade-offs among production, energy, and economic indices in a flexible manufacturing environment is of substantial practical importance.

(iii) In terms of scheduling methodology, most studies have employed mixed-integer linear programming or metaheuristic algorithms, which require complex mathematical modeling and are constrained by factors such as problem size and solution time. Only very few consider the new solution approaches using data-driven and model-free methods.

3. Problem formulation

With the growing consciousness of sustainable manufacturing, a variety of models featuring distinct constraints and objectives have been created for energy-efficient scheduling in flexible job shops. Drawing on (Wang et al., 2020), this study focuses on a flexible manufacturing setting composed of n jobs and m machines, where each job J_i consists of a sequence of operations $\left\{O_{i1}, O_{i2}, \ldots, O_{in_i}\right\}$. Every operation O_{ij} can be processed by a specific subset $M_{ij} \subseteq M$ of eligible machines. To streamline the model, we assume that the workshop's energy consumption only entails power usage from machines, which is divided into process and standby energy categories. The energy consumption for operation O_{ij} is influenced by the processing time t_{ijk} and power p_{ijk}, with these parameters varying among the candidate machines. A summary of the symbols used to model the problem and their corresponding meanings is presented in Appendix A.

3.1. IBDR under study

We investigate a day-ahead invitation-based IBDR mechanism described as follows:

Under specific conditions such as anticipated generating resource shortages or extreme temperatures exceeding set thresholds (Li & Hong,

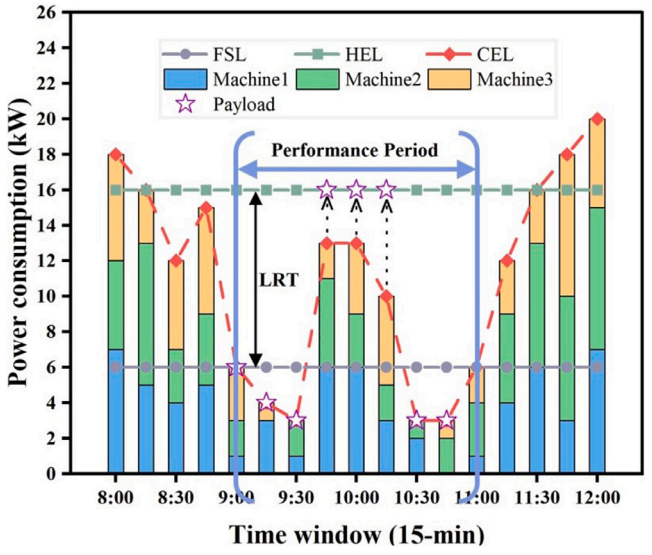


Fig. 1. Example of the proposed IBDR programs.

2016), electric utilities issue Day-ahead peak shaving invitations (DA-PSI) to industrial consumers. Three critical parameters define these invitations: (1)***Response Interval***: This is the duration of the DR event, typically divided into several equal sub-time windows to monitor load fluctuations more precisely. For example, a performance period beginning at T_s and extending through L subwindows is represented by a continuous interval set $\Omega = \{T_s, T_s + \Delta t, \ldots, T_s + L\Delta t\}$, with each subwindow lasting $\Delta t = 15$ minutes in this paper. (2)***Reduction Requirements*** (Firm service level, FSL (see Perraudat et al., 2022)): This involves a maximum power load limit during the performance period, which is the difference between the historical electric load (HEL) measured before IBDR subscription and the reduction threshold (LRT) for the current invitation. (3)***Incentive Rate*** I_{rate}: If the manufacturing workshop accepts the DA-PSI and complies with the reduction requirements during the performance period, it will earn monetary rewards. In this research, the incentive rate is denominated in CNY/kW.

The IBDR mechanism introduced in this study demonstrates enhanced flexibility compared to the methods discussed in Perraudat et al. (2022). Specifically, by easing reduction demands from the entire performance period to individual subtimeframes and eliminating penalties, this mechanism broadens the participation opportunities for small to medium manufacturing enterprises and energy consumers. Fig. 1 depicts the incentive program for a HEL of 16 kW with an LRT of 10 kW. The performance period commences at 9:00 AM and extends for 2 h ($L = 8$). Actual power consumption and corresponding payload for each subwindow during the performance period are distinctly marked in the diagram. By analyzing these markings, it is straightforward to determine that the TPL value of the current electric load (CEL) curve (illustrated by the red line) is $\sum[6, 4, 3, 16, 16, 16, 3, 3] = 67$ kW. At an incentive rate of 2 CNY/kW, participants are eligible to earn a monetary reward of $(8 \cdot 16 - 67) \cdot 2 = 122$ CNY.

3.2. Objective function

We optimize the operational metrics across three dimensions – production efficiency, energy efficiency, and economy – using the following objective function:

$$min \left\{ makespan = \max_{1 \le i \le n} C_i, TEC = \sum_{i=1}^{n} \sum_{j=1}^{n_i} \sum_{k=1}^{m_i} t_{ijk} p_{ijk} X_{ijk}, TPL = \sum_{i=1}^{n} \sum_{j=1}^{n_i} \sum_{k=1}^{m_i} \sum_{l=ps}^{pe} p_{ijk} Y_{ijkl} \right\} \quad (1)$$

Subject to:

$$S_{i,j} + \sum_{k=1}^{m_{ij}} t_{ijk} \cdot X_{ijk} \le S_{i,j+1} \forall i \in J, j \in \{1, 2, \ldots, n_i - 1\} \quad (2)$$

$$\sum_{k=1}^{m_{ij}} X_{ijk} = 1 \forall i \in J, j \in \{1, 2, \ldots, n_i\} \quad (3)$$

$$S_{i'j'} \ge S_{ij} + t_{ijk} - M \cdot (3 - Z_{iji'j'} - X_{ijk} - X_{i'j'k}) \quad \forall i, i' \in J, j, j' \in \{1, 2, \ldots, n_i\}, k \in M_{ij} \cap M_{i'j'} \quad (4)$$

$$S_{ij} \ge S_{i'j'} + t_{i'j'k} - M \cdot (2 + Z_{iji'j'} - X_{ijk} - X_{i'j'k}) \quad \forall i, i' \in J, j, j' \in \{1, 2, \ldots, n_i\}, k \in M_{ij} \cap M_{i'j'} \quad (5)$$

$$C_i \ge S_{in_i} + t_{in_ik} \cdot X_{in_ik} \forall i \in J, k \in \{1, 2, \ldots, m_{in_i}\} \quad (6)$$

$$\sum_{l=1}^{u} Y_{ijkl} = t_{ijk} \cdot X_{i,j,k} \quad \forall i \in J, \ j \in \{1, 2, \ldots, n_i\}, \ k \in \{1, 2, \ldots, m_{ij}\} \quad (7)$$

$$S_{ij}, C_i \ge 0 \quad \forall i \in J, j \in \{1, 2, \ldots, n_i\} \quad (8)$$

$$X_{ijk}, Y_{ijkl}, Z_{iji'l} \in \{0, 1\} \quad \forall i, i' \in J, j, j' \in \{1, 2, \ldots, n_i\}, k \in \{1, 2, \ldots, m_i\}, l \in \{1, 2, \ldots, u\} \quad (9)$$

where objectives are defined as Eq. (1); Constraint (2) ensures that all operations of the same job follow the specified processing precedence; Eq. (3) guarantees that only one feasible machine is assigned to each operation; Constraints (4) and (5) ensure that each operation is processed only after its predecessor on the same machine has finished and all preceding operations of the same job are completed; Constraint (6) ensures that the completion time of each job must be after completing its final operation; Eq. (7) ensures that the total processing time of each operation equals the sum of its processing times across all periods; Finally, Constraints (8) and (9) define the decision scope of the variable.

4. Methodology

4.1. Framework

To address the EFJSP-IBDR, the problem is initially decomposed into a set of subproblems based on the preference vector. These subproblems are formalized as MDPs, utilizing the HDG and four sets of raw features to represent scheduling states, and a bilayer aggregation function of trajectory and action is designed to serve as reward functions (Section 4.2). Subsequently, we develop a novel graph neural network scheduler comprising modules for representation learning and policy learning to tackle each MDP and obtain approximate Pareto-optimal solutions (Section 4.3). Moreover, a similarity-based parameter transfer method is employed to accelerate the training of scheduling strategies under various preference vectors (Section 4.4). The comprehensive framework of our method is illustrated in Fig. 2.

4.2. MDP for EFJSP-IBDR

Regarding the multi-objective dimension, drawing on the approach of using DRL to solve the multi-objective traveling salesman problem (Li et al., 2020), we decompose the EFJSP-IBDR into N scalar optimization subproblems. Each subproblem is characterized by a specific preference vector $\lambda^b, b \in [1, 2, \ldots, N]$, for example, $(1, 0), (0.8, 0.2), \ldots, (0, 1)$ for a biobjective problem, and is modeled via an MDP.

State: In this study, we improve the HDG from (Song et al., 2022) to represent the scheduling state s_t at each time step t (the timestamp at

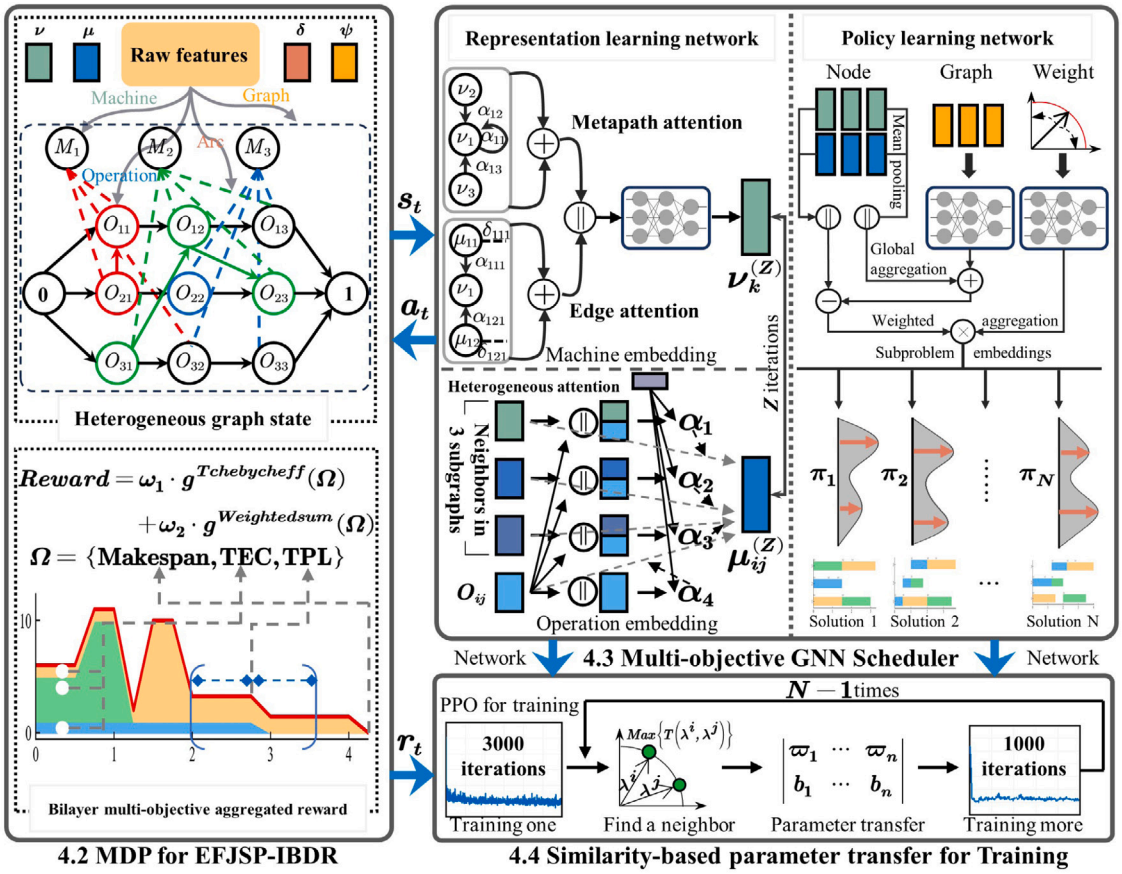


Fig. 2. Method framework.

which at least one operation is allocable). The HDG proposed in Song et al. (2022) fails to represent the processing order of operations on the same machine, thus limiting the detailed portrayal of the scheduling scheme. To resolve this, we design an improved HDG (IHDG) by introducing directed machine processing order arcs. Furthermore, to enhance the visual clarity of the IHDG and facilitate future feature extraction, we partition it into three subgraphs as illustrated in Fig. 3: 3(a) depicts the IHDG; 3(b) details the process constraint subgraph, which expresses the sequential relationship of operations; 3(c) represents the machining capability subgraph, showing the processable relationships between operations and machines and 3(d) outlines the processing queue subgraph, detailing the order of scheduled operations on the same machine.

Additionally, we develop four new sets of raw features for each component within the IHDG to reflect their states time step t. As illustrated in Appendix B, these features are derived from two perspectives: the topological structure of the IHDG and the characteristics of production and energy perspectives in the EFJSP-IBDR. Notably, compared to Song et al. (2022), we add a new set of vectors termed IBDR features. These features reflect the electricity consumption context of the current scheduling scheme during the performance period, which cannot be conveyed through individual nodes or arcs. Thus, we categorize these as graph-level features during the feature extraction process.

Action: We define a feasible operation-machine combination ($O_{ij} - M_k$) as an action $a_t \in A_t$ at time step t, where O_{ij} is eligible (the preceding process has been completed) and M_k has the capacity to process O_{ij}. The action set A_t is step dependent and the number of its elements does not exceed $n \times m$. The advantage of this design is the ability to search for solutions directly.

Reward: When designing the reward function, we focus on two dimensions: multi-level and multi-objective. For the multi-level aspect, inspired by the concepts from (Liu et al., 2023; Qin et al., 2023), we establish a bilayer reward structure for trajectories and actions. The trajectory reward, denoted as r_{tra}, corresponds to the actual value of the optimization target, guiding the overarching optimization strategy. Given that r_{tra} can only be accurately determined post-scheduling, we utilize the difference in expected optimization objectives at adjacent time steps as action reward r_{act} to assess each action's contribution. Furthermore, in terms of redistributing trajectory rewards, while (Liu et al., 2023) suggests allocating the reward solely to the final time step, (Qin et al., 2023) advocates for an equal distribution across all scheduling actions. Considering our dynamic action space, we devise a method to dynamically allocate r_{tra} based on $|A_t|$ at each time step. This approach guarantees that time steps with greater decision complexity (a larger number of $O_{ij} - M_k$) are awarded higher rewards. For the multi-objective aspect, taking cues from (Zheng et al., 2018), we craft a weighted hybrid multi-objective aggregated reward function based on the preference vector λ^b for each subproblem. Specifically, r_{act} is determined using the Tchebycheff method, while r_{tra} employs the weighted sum approach. For subproblem b, the reward r_t at time step t is defined as:

$$\begin{aligned} r_t(\lambda^b) &= r^t_{tra}(\lambda^b) + r^t_{act}(\lambda^b) \\ r^t_{tra}(\lambda^b) &= \frac{|A_t|}{\sum_{t=0}^{|\mathcal{O}|}|A_t|}\sum_{g=1}^{objs}\lambda^b_g f_g(|\mathcal{O}|) \\ r^t_{act}(\lambda^b) &= TE_{t-1}(\lambda^b) - TE_t(\lambda^b) \\ TE_t(\lambda^b t) &= \max_{1\leqslant g\leqslant objs}\left\{\lambda^b_g\left|f_g(t) - z^*_g\right|\right\} \end{aligned} \tag{10}$$

where: $objs$ specifies the number of optimization objectives (for this research, $objs = 3$); λ^b_g denotes the weight of the g-th objective in vector λ^b; $f_g(t)$ indicates the value of the g-th objective at time step

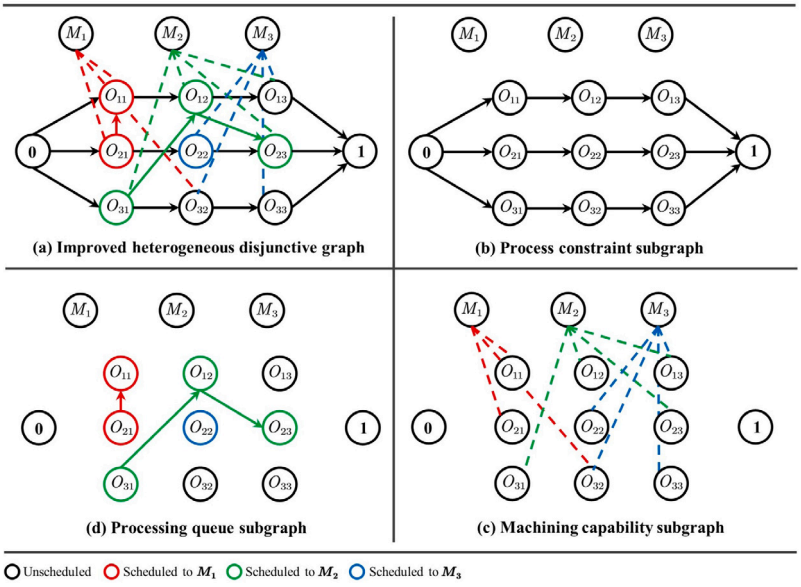


Fig. 3. The IHDG and disassembled subgraphs.

t; z_g^* signifies the reference value of the g-th objective, which is the minimum value encountered during training; $\mathcal{O}$ represents the set of operation nodes in the IHDG.

Transition and policy: Based on s_t and a_t, the scheduling environment deterministically transits to a new state s_{t+1}. For each s_t, the policy $\pi_\theta(a_t|s_t)$ defines a probability distribution of all actions over A_t (Rui et al., 2024). The aim of GRL is to refine the parameter θ via training perpetually.

4.3. Multi-objective GNN scheduler

In adherence to the research framework established in Rui et al. (2024), we design a multi-objective version of the graph neural network scheduler to extract embedded information and select scheduling actions. The redesigned scheduler comprises both a representation learning network and a policy learning network. For the representation learning network, this study introduces a dual attention mechanism to embed operation nodes and machine nodes separately. The policy learning network focuses on aggregating features extracted by the representation learning network with the weight vectors of subproblems. This aggregation process results in comprehensive subproblem embeddings suitable for policy learning.

4.3.1. Dual attention mechanism-based representation learning network

Machine node embedding via metapath-edge attention: This study introduces a metapath-edge attention mechanism to enhance feature extraction of machine nodes. Compared to the embedding scheme described in Rui et al. (2024), this approach includes two significant advancements: (1) *Neighbors definition*: Whereas previous research only considered graph topological neighbors, the current study expands the definition to encompass semantic neighbors of machine nodes. (2) *Edge features integration*: Instead of merely concatenating edge features with node features, this work utilizes an edge attention method to integrate these features more effectively.

In the context of metapath neighbors, based on research (Wang et al., 2019) in social networks, we define the metapath as a pathway comprising disjunctive arcs with two distinct machine nodes that connect to a single operation node within the machining capability subgraph. As shown in Fig. 4, machine node M_k and all other machine nodes connected by a metapath to M_k are deemed its metapath neighbors, denoted as $\mathcal{N}_t^{\Phi}(M_k)$. From the scheduling perspective, metapath serves to reflect the competitive relationships among compatible machines vying for the same operations. Moreover, we utilize Graph attention networks (GAT) to embed the neighbors based on metapath (Veličković et al., 2017), with the detailed process outlined below:

$$\begin{aligned} v_k^* &= \text{MetapathGAT}(v_k) = \Theta_1 \cdot v_k + \Big(\sum_{c \in \mathcal{N}_t^{\Phi}(M_k)} \alpha_{ck}(\Theta_1 \cdot v_c) \Big) \\ \alpha_{ck} &= \text{softmax}(\text{LeakyReLU}(a_0^T [\Theta_1 \cdot v_k \parallel \Theta_2 \cdot \Sigma_{ij \in \Phi(ck)} \mu_{ij}])) \end{aligned} \tag{11}$$

where: v_k and v_c denotes the feature vector of machine node M_k and M_c before embedding; Θ_1 represents the learnable weight matrix used to transform the features of the machine node into a designated embedding dimension; a_0 represents the learnable weight matrix; LeakyReLU (Veličković et al., 2017) denotes the activation function; softmax stands for the normalization of all metapath neighbors; Θ_2 indicates another learnable weight matrix for transforming the features of operation nodes; $\parallel$ symbolizes the cascading operator; μ_{ij} is the feature vector for the operation node O_{ij}. Moreover, $\Phi(ck)$ signifies the collection of operation nodes that have a metapath connection with both machine nodes M_c and M_k. By summing up the features of all operation nodes competing between M_c and M_k as the features of metapath neighbors, thus allowing machine nodes with higher homogeneity (i.e., similar processing capabilities) to have larger assigned attention coefficients.

In terms of edge features processing, the disjunctive arcs connecting operation and machine nodes carry information about the processing time and power for operation-machine pairs, which is crucial for scheduling decisions. Given that node feature dimensions greatly exceed those of edge features, directly concatenating them could result in a diminishing influence of edge features on the final embedding outcome over successive iterations. To tackle this issue, we utilize the EdgeGAT (Monninger et al., 2023) to incorporate edge features into the processes of message passing and neighborhood aggregation. The process of embedding graph topological neighbors with edge features is depicted as:

$$\begin{aligned} v_k^{\diamond} &= \text{EdgeGAT}(v_k) = \sum_{ij \in \mathcal{N}_t(M_k)} \alpha_{ijk} \cdot (\Theta_2 \cdot \mu_{ij} + \Theta_3 \cdot \delta_{ijk}) \\ \alpha_{ijk} &= \text{softmax}(\text{LeakyReLU}(a_0^T [\Theta_1 \cdot v_k \| \Theta_2 \cdot \mu_{ij} \| \Theta_3 \cdot \delta_{ijk}])) \end{aligned} \tag{12}$$

where: $\mathcal{N}_t(M_k)$ represents the collection of neighboring operation nodes at time step t; Θ_3 denotes the learnable weight matrix that

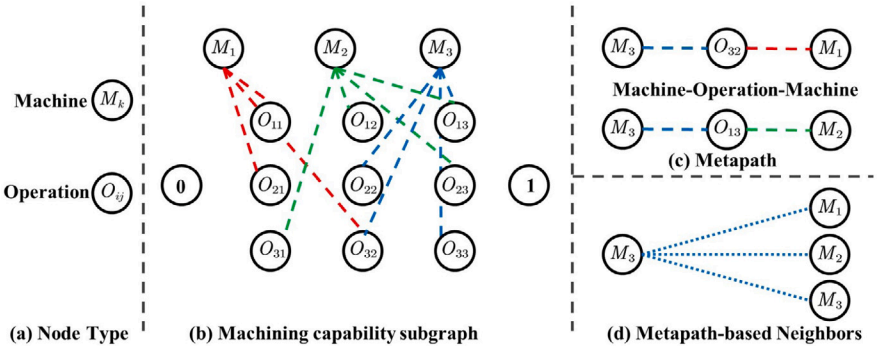


Fig. 4. Illustration of metapath neighbors for machine nodes.

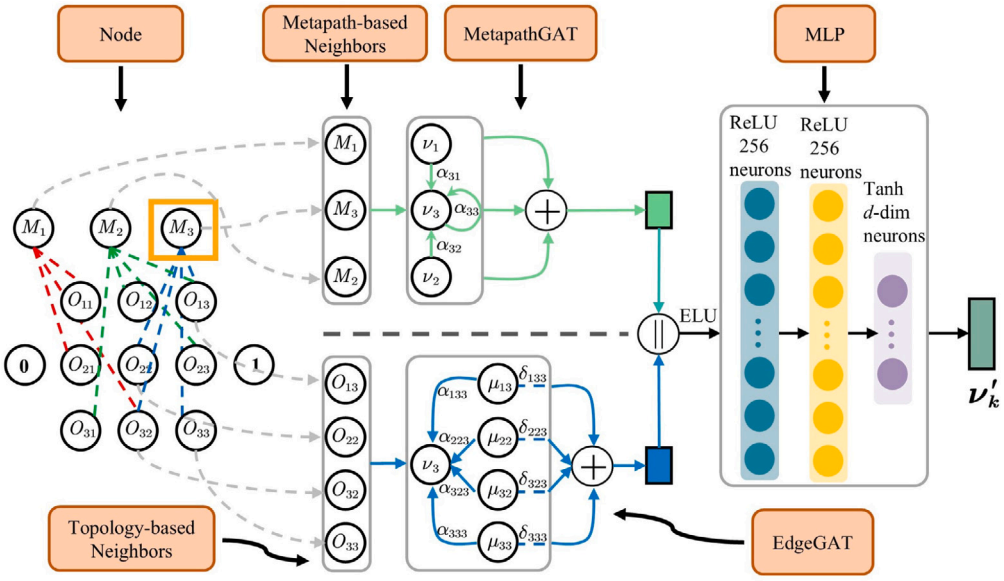


Fig. 5. Embedding scheme for machine nodes.

transforms the dimensions of edge features; δ_{ijk} refers to the edge feature vector.

As illustrated in Fig. 5, MetapathGAT is capable of transmitting messages from semantic neighbors, whereas EdgeGAT is able to gather information from various edges. By integrating these two mechanisms, the feature representation capabilities of machine nodes are enhanced, which is advantageous for subsequent decision-making and optimization processes. The final embedding outcome can be denoted as $v'_k = \mathrm{MLP}(\mathrm{ELU}(v^*_k \parallel v^\diamond_k))$. Here, the MLP consists of a $2d$-dimensional input layer, an d-dimensional output layer, and two d_h-dimensional hidden layers, with the ELU function (Rui et al., 2024) used for activation in these hidden layers.

Heterogeneous attention-based operation node embedding: For the IHDG designed in this study, we enhance the Heterogeneous attention mechanism (HAM) described in our prior study (Rui et al., 2024) to extract features of operation nodes effectively. Regarding operation node O_{ij}, the subgraphs of process constraints and processing queues delineate five categories of neighboring operation nodes: itself O_{ij}, process predecessors $O_{i,j-1}$, process successors $O_{i,j+1}$, queue predecessors $O_{i',j'}$, and queue successors $O_{i'',j''}$. The encoding outcomes for these categories are represented sequentially as $f_1 = \mathrm{MLP}(\mu_{ij})$, $f_2 = \mathrm{MLP}(\mu_{i,j-1})$, $f_3 = \mathrm{MLP}(\mu_{i,j+1})$, $f_4 = \mathrm{MLP}(\mu_{i',j'})$, and $f_5 = \mathrm{MLP}(\mu_{i'',j''})$. For the neighbor machine nodes defined by the machining capability subgraph, the method outlined in our previous research continues to be applied, with encoding outcomes illustrated below:

$$f_6 = \frac{\Sigma_{k\in\mathcal{N}_t(O_{ij})}\mathrm{MLP}(v_k)}{|\mathcal{N}_t(O_{ij})|} \tag{13}$$

where: $\mathcal{N}_t(O_{ij})$ denotes the set of neighboring machine nodes at time step t. Ultimately, the encoding results of the aforementioned neighboring nodes are aggregated through an attention mechanism, culminating in the final embedding of the operation node O_{ij}, as illustrated below:

$$\begin{aligned} \mu'_{ij} &= \sum_{f_x\in\mathcal{F}} \alpha^{1,x} f_x \\ \alpha^{1,x} &= \frac{\exp\left\{\mathrm{LeakyReLU}(a_1^T[f_x \parallel f_1])\right\}}{\sum_{f_y\in\mathcal{F}}\exp\left\{\mathrm{LeakyReLU}(a_1^T[f_y \parallel f_1])\right\}} \end{aligned} \tag{14}$$

where: $\mathcal{F} = \{f_1 \cup f_2 \cup f_3 \cup f_4 \cup f_5 \cup f_6\}$; $a_1 \in \mathbb{R}^{2d}$ signifies the learnable attention parameter matrix. The complete embedding process for operation nodes is depicted in Fig. 6.

The attention modules described above are sequentially linked to form an embedding layer. This layer transforms node-level features μ_{ij} and v_k in the graph to d-dimensional embeddings via Z iterations of stacking and average pooling. Moreover, for graph-level feature ψ, feature extraction is performed using an MLP that includes

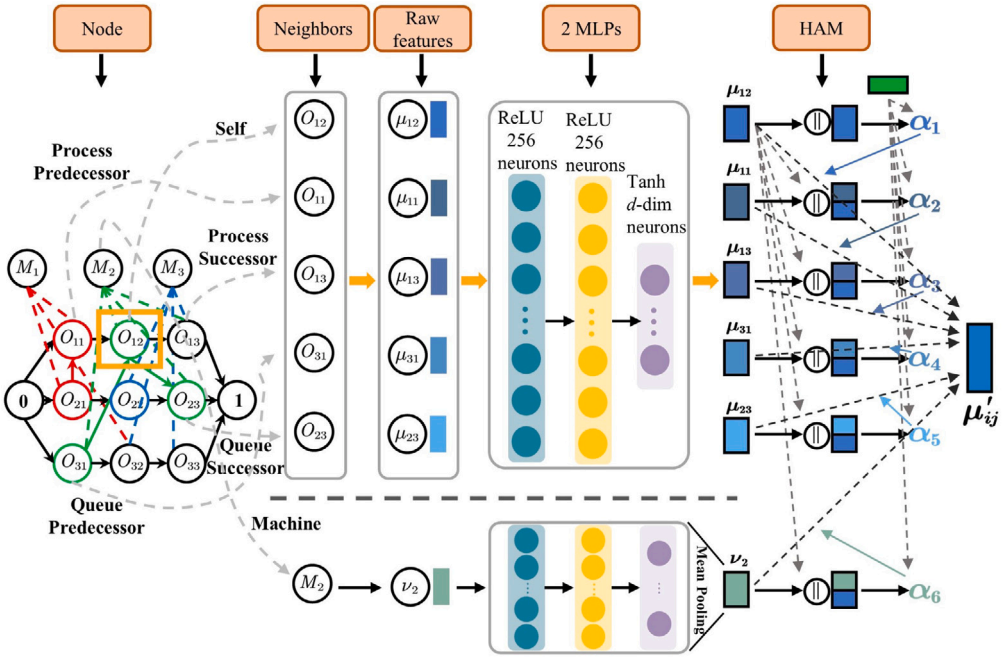


Fig. 6. Embedding scheme for operation nodes.

a 3-dimensional input layer, an $2d$-dimensional output layer, two d_h-dimensional hidden layers, and ReLU activation. Ultimately, the computed outcomes are combined with the node-level embeddings to produce the complete graph embedding, as detailed below:

$$h_G^t = \left[\left(\frac{1}{|\mathcal{O}|}\sum_{O_{ij}\in\mathcal{O}} \mu_{ij}^{'(Z)}\right) \| \left(\frac{1}{|M|}\sum_{M_k\in M} v_k^{'(Z)}\right)\right] + \mathrm{MLP}(\psi) \tag{15}$$

where: M is the set of machines; $\mu_{ij}^{'(Z)}$ and $v_k^{'(Z)}$ are in turn the feature embeddings of O_{ij} and M_k after Z iterations.

4.3.2. Policy learning network with decomposition

As outlined before, our research decomposes the EFJSP-IBDR into N distinct subproblems, which are addressed through various policy learning networks (identical structures but with differing network parameters). To assist the networks in generating suitable strategies based on input preferences, we initially develop a preference network to encode the correlated weight vectors. According to Wang et al. (2023), this preference network comprises a single linear layer and a ReLU activation layer. The encoding process can be represented as $\Lambda_b = \mathrm{ReLU}(\mathrm{Linear}(\lambda^b))$, which involves a linear layer with an input dimension of 3 and an output dimension of $2d$. Following this, the feature embeddings and preference codings are aggregated through matrix multiplication, and the results are funneled into the Actor and Critic networks for strategy development. In the end, the input for the Actor network is denoted as $\mathrm{Actor} = [[\mu_{ij}^{'(Z)} \| v_k^{'(Z)}] - h_G^t] \cdot \Lambda_b$, and for the Critic network as $\mathrm{Critic} = h_G^t \cdot \Lambda_b$.

The design rationale behind this is that the Actor network requires identifying the relative advantage of the current action by deriving the feature extraction result from subtracting the graph embedding from the concatenated embeddings of operation and machine nodes. In contrast, the Critic network focuses on scoring the current state and thus directly utilizes the graph embedding as the operator.

4.4. Similarity-based parameter transfer for training

Since each subproblem is solved by a neural network, a key issue during the training phase is how to reduce training time while maintaining model accuracy. Drawing on the findings from Li et al. (2020), we introduce a similarity-based parameter transfer strategy for training the neural networks associated with each subproblem. Notably, two subproblems with adjacent weight vectors may have nearly identical optimal solutions, as suggested by Zhang and Li (2007), facilitating the use of knowledge transfer from a neighboring subproblem to aid resolution. Diverging from (Li et al., 2020), our study quantifies the neighboring relationships between subproblems through vector similarity coefficients. The Tanimoto coefficient, an extension of angle-based cosine similarity, excels at handling high-dimensional sparse data and promotes efficient computation (Wang et al., 2024; Zhao et al., 2023). Given the high-dimensional, time-slot-related variables of EFJSP-IBDR and the need to shorten training time, we introduce the Tanimoto coefficient to efficiently measure the statistical correlation among different subproblems. The weight vector with the highest Tanimoto coefficient is designated as the neighbor. In a tri-objective optimization scenario, the Tanimoto coefficient between two weight vectors, $\lambda^1 = (x_1, x_2, x_3)$ and $\lambda^2 = (y_1, y_2, y_3)$, is depicted as:

$$T\left(\lambda^1, \lambda^2\right) = \frac{\sum_{i=1}^{3}\left(\lambda_i^1 \times \lambda_i^2\right)}{\sum_{i=1}^{3}\left(\lambda_i^1\right)^2 + \sum_{i=1}^{3}\left(\lambda_i^2\right)^2 - \sum_{i=1}^{3}\left(\lambda_i^1 \times \lambda_i^2\right)} \tag{16}$$

If subproblem b is resolved, we utilize its parameters to initiate training for the unsolved subproblem $\hat{b}$, which has the smallest Tanimoto coefficient, thereby reducing the number of training iterations required. However, the initial subproblem lacks auxiliary referential knowledge, necessitating comprehensive parameter training.

The training of the subproblem models continues to utilize PPO, as detailed in our prior research (Rui et al., 2024). To further expedite the model training duration, the representation learning network is trained only once and it is subsequently excluded from the parameter transfer process. The rationale for this design is that the feature extraction process does not directly contribute to the optimization objectives of the EFJSP-IBDR. Instead, it focuses on converting all state features into vectors and minimizing information loss, as described in Chen et al. (2022). The process of training based on similarity-based parameter transfer is outlined in Algorithm 1.

5. Experimental results

5.1. Configurations

We refine DA-PSI parameters and subsequently develop EFJSP-IBDR instances for both training and validation purposes. Specifically,

Algorithm 1: Similarity-based training framework for multi-objective graph neural network schedulers.

Input: The model of subproblems $\Gamma = \{[w_1, b_1], [w_2, b_2], \cdots, [w_N, b_N]\}$, weight vectors $\lambda^1, \lambda^2, \cdots \lambda^N$.

Output: The optimal model $\Gamma^* = \{[w_1^*, b_1^*], [w_2^*, b_2^*], \cdots, [w_N^*, b_N^*]\}$.

Initialize an untrained weight vector set $\Pi = \{\lambda^1, \lambda^2, \cdots \lambda^N\}$ and training count $count = 0$;
Generate random integer $num \in \{1, 2, \cdots, N\}$;
$[w_{\lambda^{num}}, b_{\lambda^{num}}] \leftarrow Random_Initialize$;
$[w^*_{\lambda^{num}}, b^*_{\lambda^{num}}] \leftarrow PPO([w_{\lambda^{num}}, b_{\lambda^{num}}], R(\lambda^{num}))$;
Remove λ^{num} from Π;
$count = count + 1$;
while $count < N$ **do**
  $\max \leftarrow \arg\max_{\lambda^{item} \in \Pi} T(\lambda^{num}, \lambda^{item})$;
  $[w_{\lambda^{\max}}, b_{\lambda^{\max}}] \leftarrow [w^*_{\lambda^{num}}, b^*_{\lambda^{num}}]$;
  $[w^*_{\lambda^{\max}}, b^*_{\lambda^{\max}}] \leftarrow PPO([w_{\lambda^{\max}}, b_{\lambda^{\max}}], R(\lambda^{\max}))$;
  $num \leftarrow \max$;
  Remove λ^{num} from Π;
  $count = count + 1$;
Given inputs of the EFJS-IBDR, the Pareto frontier can be directly calculated by Γ^*.

considering the dynamic nature of the DA-PSI time windows and a fixed production cycle of 24 h, we posit that the start time and duration of the performance period adhere to uniform distributions $U(8, 19)$ and $U(2, 4)$, respectively. To simplify the dataset, we set the HEL as the product of the number of machines in the instance and the average possible processing power of all machines. Furthermore, LRT is indirectly depicted through the percentage reduction in HEL, with the ratio fulfilling $U(40, 60)$.

We conduct a sensitivity analysis of several key hyperparameters in Appendix C. According to the analysis, we configure the number of iterations for the representation learning network at $Z = 2$, and establish the dimensions for all MLP hidden layers at $d_h = 256$. During the learning process, the learning rate is maintained at $lr = 2.5 \times 10^{-4}$, and the discount factor at $\gamma = 0.96$. Following the method for generating weight vectors described in Zhang and Li (2007), we divide each objective dimension into 16 segments and define $N = 153$ subproblems. For PPO parameters, parameters are set for clipping parameter $\epsilon = 0.2$, policy loss coefficient $c_1 = 1$, value function coefficient $c_2 = 0.5$ and entropy coefficient $c_3 = 0.01$ based on the recommendations from (Hill et al., 2018).

Fig. 7(a) displays the average rewards and losses per iteration collected during the training of the first subproblem derived from random sampling while Fig. 7(b) illustrates the training details for the new subproblem created through parameter transfer from the fully trained model. It is evident that with the increase in training iterations, both the fully trained subproblem and those trained following parameter transfer converge well. All experiments are conducted on a device equipped with an Intel Xeon(R) Silver 4210R CPU, four Nvidia GeForce RTX 3090 GPUs, and running Ubuntu 22.04 64-bit OS. Additionally, the approach presented in this paper is developed with PyTorch and Python 3.12, with the code being openly accessible.[2]

5.2. Comparisons between DSGRL and Gurobi

In order to verify the proposed model's ability to find optimal solutions, we compare DSGRL (GRL with dual attention mechanism and similarity-based parameter transfer strategy) and Gurobi (version 10.0.3) on three Kacem (Kacem et al., 2002) instances (Kacem01-03). We set the processing power based on (Wang et al., 2020) and define the corresponding IBDR parameters as follows: a performance period of $[4, 12]$, an FSL of m and an HEL of $4m$, where m is the number of machines in each instance. Since Gurobi is unable to solve multi-objective optimization problems directly, the weighted sum method is employed to create four single-objective optimization subproblems for each instance by selecting four sets of weights, namely $(0.8, 0.1, 0.1)$, $(0.4, 0.3, 0.3)$, $(0.1, 0.8, 0.1)$ and $(0.1, 0.1, 0.8)$. These weights represent different preferences for the three objectives. We set Gurobi's maximum running time to 3600s and record the best solution if it cannot find the optimal solution within that time. The results of the comparison between DSGRL and Gurobi are shown in Table 1, where Gap is obtained from $100 \times \frac{\text{Gurobi-DSGRL}}{\text{Gurobi}}$. Based on the results, Gurobi finds optimal solutions for all subproblems of Kacem01 ($n \times m = 4 \times 5$). However, it struggles with the larger Kacem02 ($n \times m = 8 \times 8$) and Kacem03 ($n \times m = 10 \times 7$). In contrast, DSGRL solves 11 out of 12 subproblems successfully while also requiring less computational time. These findings indicate that using Gurobi alone to solve EFJSP-IBDR is feasible but not efficient, especially for large-scale instances. Therefore, advanced algorithms remain necessary for this problem.

Table 1
Comparison results of DSGRL and Gurobi.

Instances	Weight	Gurobi		DSGRL		Gap
		Value	Time/s	Value	Time/s	
Kacem01	(0.8,0.1,0.1)	22.46	167.3	22.46	0.72	0
	(0.4,0.3,0.3)	28.28	32.43	28.28	0.66	0
	(0.1,0.8,0.1)	43.38	45.87	43.38	0.59	0
	(0.1,0.1,0.8)	12.29	59.22	12.85	0.48	−4.56
Kacem02	(0.8,0.1,0.1)	43.24	3600	36.16	1.53	16.37
	(0.4,0.3,0.3)	52.85	3600	45.19	1.51	14.49
	(0.1,0.8,0.1)	76.38	3600	63.64	1.31	16.68
	(0.1,0.1,0.8)	19.80	3600	18.93	1.30	4.39
Kacem03	(0.8,0.1,0.1)	50.67	3600	39.09	10.36	22.85
	(0.4,0.3,0.3)	55.03	3600	47.81	9.39	13.12
	(0.1,0.8,0.1)	78.75	3600	66.95	9.67	14.98
	(0.1,0.1,0.8)	21.29	3600	19.41	9.73	8.83

5.3. Effectiveness of each improvement of the DSGRL

To verify the effectiveness of each modified components proposed in this study, we performed ablation experiments. GRL-ND indicates the feature extraction process employs the simple MLP architecture from our prior work (Rui et al., 2024), while the training process utilizes the similarity-based parameter transfer strategy. GRL-NS replaces the similarity-based parameter transfer with the original parameter transfer strategy proposed in Li et al. (2020), while preserving the corresponding dual attention mechanism. We compare these two variants with DSGRL in the MK (MK01-10) instances (Brandimarte, 1993). The performance period parameter is set to $[20, 30]$, and the remaining energy-related parameters are consistent with Section 5.2.

In terms of evaluation criteria, this research utilizes Hypervolume (HV) and Spacing to assess the performance of multi-objective optimization algorithms from different perspectives. HV quantifies the volume of the objective space enclosed by the non-dominated solution set obtained by the algorithm and the reference point. A larger HV value indicates better diversity and convergence of solutions. We determine the reference point by shifting the maximum values on each objective of the non-dominated vectors from the approximation sets of all algorithms by a set offset of 5%. Spacing measures the standard deviation of the minimum Euclidean distance between each solution in the non-dominated solution set and other solutions. A lower Spacing value denotes improved uniformity of solutions. Table 2 list

[2] https://github.com/ruizhangjie/EFJSP-IBDR-GRL

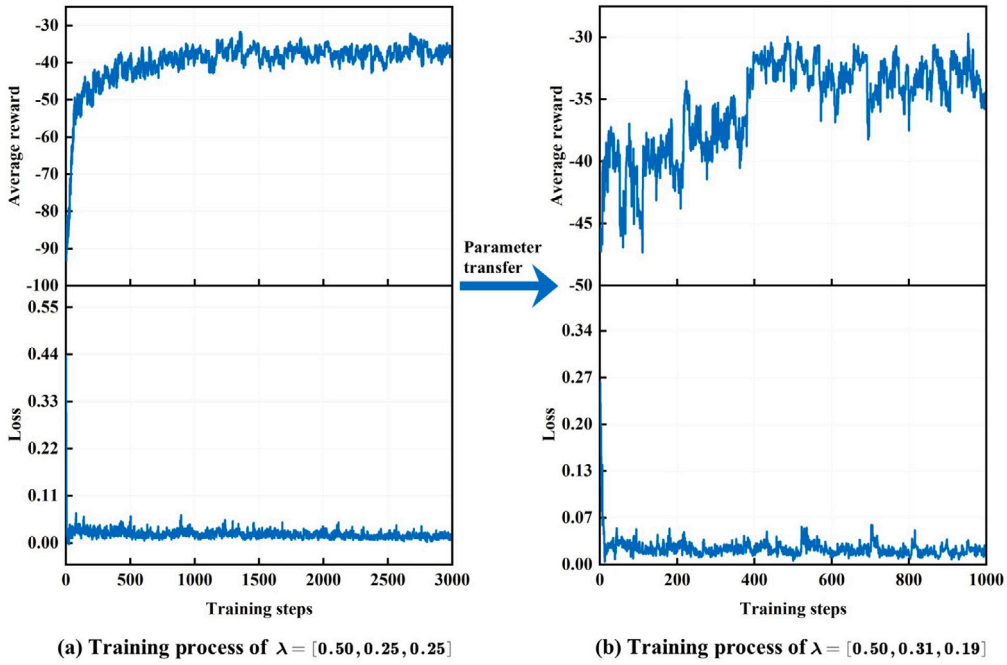


Fig. 7. Training curves for two sample subproblems based on parameter transfer.

Table 2
Comparison results of DSGRL and its variants.

Instances	HV			Spacing		
	GRL-ND	GRL-NS	DSGRL	GRL-ND	GRL-NS	DSGRL
MK01	1.22e+07	1.29e+07	**1.32e+07**	16.21	17.94	**11.23**
MK02	6.58e+06	6.63e+06	**6.72e+06**	16.10	16.07	**13.46**
MK03	2.25e+08	2.31e+08	**2.48e+08**	18.89	21.02	**18.47**
MK04	2.06e+07	2.11e+07	**2.17e+07**	20.20	21.22	**18.96**
MK05	7.37e+05	7.45e+05	**7.70e+05**	4.91	4.06	**3.75**
MK06	2.35e+08	2.50e+08	**2.67e+08**	38.20	38.77	**33.08**
MK07	6.24e+07	6.28e+07	**6.41e+07**	17.01	16.17	**15.16**
MK08	5.55e+05	5.78e+05	**6.04e+05**	3.41	4.23	**3.56**
MK09	6.02e+06	6.16e+06	**7.84e+06**	4.40	7.48	**4.22**
MK10	1.65e+07	1.78e+07	**2.14e+07**	6.68	7.89	**6.10**
better	0	0	**10**	0	0	**10**

the compared results, and the best value obtained for each instance is marked with bold.

As shown in Table 2, DSGRL is significantly better than the other variants in every instance for both the HV and Spacing metrics. In terms of HV, GRL-ND performs the worst in all cases. This finding suggests that the dual attention-based scheduling feature extraction method greatly boosts overall algorithm performance compared to simple cascading after linear transformation. Additionally, DSGRL demonstrates greater advantages as the instance size increases. Regarding Spacing, DSGRL is more competitive than GRL-NS. Therefore, spatial similarity can be adopted for the training phase domain. The introduction of spatial similarity as the basis for selecting neighborhood preference vectors effectively ensures the good distribution and convergence of the Pareto solution set. Hence, these results prove that all the improved strategies are effective for enhancing the performance of DSGRL.

5.4. Performance analysis

In order to validate the performance of the proposed algorithm, we compare DSGRL with MOEA/D (Zhang & Li, 2007) and four other state-of-the-art algorithms. MOEA/D uses Tchebycheff decomposition to aggregate weights, which is used as a baseline method. The four additional methods are BEG-NSGA-II (Deng et al., 2017), INSBBO (An et al., 2021), MODJA (Caldeira & Gnanavelbabu, 2021) and PGRL (Su et al., 2024). All of these methods achieve good results on multi-objective FJSP. The parameter settings are severely by the details of the original article to achieve a reported performance. Meanwhile, we introduced larger-scale DP (DP01a-18a) instances (Dauzère-Pérès, 1997) and set the corresponding performance period parameter to $[40, 60]$. All algorithms are run independently for 20 times, and the average values of the HV, Spacing and Time metrics are list in Tables 3–5.

The comparison results demonstrate the advancement of our proposed method. According to the HV results in Table 3, DSGRL achieved 20 optimal results out of 28 instances. It also outperforms the classical evolutionary algorithms (MOEA/D and BEG-NSGA-II) in 25 instances. Compared to the state-of-the-art metaheuristic algorithms, DSGRL shows advantages on large-scale problems (DP10a-18a) while yields better results in all instances than the latest multi-objective GRL method. This improvement is primarily attributed to the incorporation of metapath-edge attention rather than a simple linear transformation in the machine node embedding. Regarding the Spacing results, Table 4 indicates that DSGRL achieved 21 optimal results. It outperforms BEG-NSGA-II and PGRL across most instances. In comparison with INSBBO and MODJA, DSGRL shows a slight disadvantage in some small-scale instances (e.g., MK01, MK05, DP02a and DP03a). As for the Time metric, Table 5 shows that the two GRL-based methods exhibit a significant advantage over search-based methods for all instances. They benefit from learning a scheduling strategy through training, which enables rapid solutions for new instances. In contrast, search-based methods must repeat the initialization and iterative evolution process for each case. Additionally, DSGRL requires more computation time than PGRL in 13 instances. However, such millisecond-level differences are imperceptible in practical industrial environments. Overall, the strategies learned by our approach generalize efficiently to out-of-distribution instances. The optimization results is not significantly inferior to state-of-the-art multi-objective metaheuristic algorithms in large-scale examples. Meanwhile, they remain competitive with the other approach of the same type.

To further demonstrate the performance of our proposed method under different IBDR parameters, we test three sets of performance period parameters: $[5, 10]$, $[20, 40]$, and $[60, 100]$, as well as three sets

Table 3
Comparison results of different algorithms on HV.

Instances	MOEA/D	BEG-NSGA-II	INSBBO	MODJA	PGRL	DSGRL
MK01	1.8911e+07	1.8544e+07	1.9171e+07	**1.9203e+07**	1.8368e+07	1.7999e+07
MK02	2.0247e+07	2.3538e+07	2.3630e+07	2.3690e+07	2.2563e+07	**2.3724e+07**
MK03	2.6492e+08	2.7376e+08	2.7662e+08	2.7409e+08	2.7133e+08	**2.7783e+08**
MK04	2.1765e+07	2.7968e+07	2.8568e+07	2.8024e+07	2.6806e+07	**2.8585e+07**
MK05	1.9489e+06	1.9411e+06	**2.0211e+06**	2.0176e+06	1.9241e+06	1.9290e+06
MK06	3.5817e+08	3.6240e+08	3.7651e+08	3.7528e+08	3.6157e+08	**3.7840e+08**
MK07	1.1377e+08	1.1343e+08	**1.1562e+08**	1.1473e+08	1.0807e+08	1.1157e+08
MK08	1.6673e+06	2.2945e+06	2.3165e+06	2.3639e+06	2.2970e+06	**2.3798e+06**
MK09	1.1512e+07	1.1535e+07	1.1903e+07	1.1910e+07	1.1200e+07	**1.1918e+07**
MK10	2.9405e+07	3.0983e+07	3.1142e+07	3.1249e+07	3.0760e+07	**3.1333e+07**
DP01a	1.3996e+05	1.5851e+05	1.6461e+05	1.6671e+05	1.5026e+05	**1.6744e+05**
DP02a	2.7561e+06	2.7643e+06	**2.8194e+06**	2.8170e+06	2.7459e+06	2.8144e+06
DP03a	1.0254e+06	1.1240e+06	1.1254e+06	1.1274e+06	1.0687e+06	**1.1321e+06**
DP04a	2.4491e+05	2.5652e+05	2.6261e+05	**2.6496e+05**	2.2781e+05	2.6249e+05
DP05a	2.1869e+06	2.3654e+06	2.3725e+06	2.3855e+06	2.2485e+06	**2.3988e+06**
DP06a	2.6741e+05	2.9874e+05	2.9894e+05	**2.9949e+05**	2.8652e+04	2.9919e+05
DP07a	1.2687e+05	1.5182e+05	1.5724e+05	**1.5734e+05**	1.3023e+05	1.5522e+05
DP08a	4.4521e+05	4.4696e+05	5.0237e+05	4.9957e+05	4.4246e+05	**5.0384e+05**
DP09a	8.4385e+06	8.6290e+06	**8.7652e+06**	8.6684e+06	8.5631e+06	8.6583e+06
DP10a	9.9025e+05	9.9259e+05	9.9550e+05	9.9424e+05	9.8937e+05	**9.9889e+05**
DP11a	2.1762e+06	2.2865e+06	2.2873e+06	2.3122e+06	2.1546e+06	**2.3390e+06**
DP12a	5.5454e+06	5.7991e+06	5.8871e+06	5.8265e+06	5.5590e+06	**5.9086e+06**
DP13a	2.6182 e+05	2.9203e+05	3.2380e+05	3.1769e+05	2.8647 e+05	**3.2413e+05**
DP14a	3.9034e+06	3.9316e+06	3.9364e+06	3.9404e+06	3.9240e+06	**3.9441e+06**
DP15a	6.7992e+07	6.7307e+07	6.8617e+07	6.8684e+07	6.8105e+07	**6.8733e+07**
DP16a	3.8319e+06	4.0258e+06	4.1390e+06	4.1319e+06	3.6910e+06	**4.1401e+06**
DP17a	2.3586e+06	2.5275e+06	2.5648e+06	2.5629e+06	2.5139e+06	**2.5700e+06**
DP18a	4.4375e+07	4.4454e+07	4.4689e+07	4.4630e+07	4.2670e+07	**4.4696e+07**
better	0	0	4	4	0	**20**

Table 4
Comparison results of different algorithms on Spacing.

Instances	MOEA/D	BEG-NSGA-II	INSBBO	MODJA	PGRL	DSGRL
MK01	13.14	17.52	**12.01**	16.39	15.92	14.20
MK02	20.63	20.89	16.44	19.03	23.88	**15.98**
MK03	22.47	19.76	23.52	18.51	23.74	**18.23**
MK04	20.07	20.28	18.32	16.97	22.06	**16.54**
MK05	4.43	4.04	3.89	**3.18**	5.33	4.62
MK06	40.86	37.02	35.51	34.93	37.93	**31.41**
MK07	19.81	16.65	15.71	14.41	15.79	**14.11**
MK08	17.87	8.46	3.93	4.54	10.04	**3.04**
MK09	10.84	9.60	7.73	5.89	11.15	**4.72**
MK10	7.88	9.25	5.92	6.80	8.65	**5.35**
DP01a	18.36	13.32	8.72	11.83	19.22	**5.69**
DP02a	9.63	5.78	**1.52**	8.51	10.28	2.02
DP03a	12.33	16.61	**5.25**	12.05	12.85	14.25
DP04a	11.80	9.09	4.86	5.87	9.73	**3.84**
DP05a	13.45	12.39	7.11	6.34	10.11	**5.28**
DP06a	12.46	14.24	9.79	**6.12**	12.20	6.80
DP07a	14.00	12.19	7.53	10.02	12.89	**5.16**
DP08a	13.40	15.42	11.24	**7.51**	13.54	7.65
DP09a	22.03	18.47	17.59	16.98	20.88	**16.91**
DP10a	10.86	10.29	8.44	5.41	12.47	**5.05**
DP11a	13.66	8.47	**5.14**	7.51	9.18	7.58
DP12a	12.72	8.08	5.74	9.52	11.44	**5.20**
DP13a	12.49	14.20	12.80	12.31	14.94	**11.84**
DP14a	10.77	6.27	5.06	5.72	9.48	**5.02**
DP15a	11.28	6.38	8.67	8.02	14.97	**5.95**
DP16a	10.31	9.44	9.12	5.26	9.30	**4.30**
DP17a	9.76	7.36	8.11	7.29	10.98	**6.31**
DP18a	11.80	10.49	7.58	7.41	15.07	**6.34**
better	0	0	4	3	0	**21**

Table 5
Comparison results of different algorithms on Time (s).

Instances	MOEA/D	BEG-NSGA-II	INSBBO	MODJA	PGRL	DSGRL
MK01	129.30	202.77	339.26	244.99	**11.96**	12.52
MK02	128.42	200.28	345.55	243.78	12.16	**12.04**
MK03	383.46	394.84	556.44	448.29	14.94	**14.94**
MK04	229.78	298.55	493.81	403.10	**11.94**	13.15
MK05	196.8	249.35	458.89	365.28	**9.66**	10.15
MK06	406.18	414.30	546.67	451.14	**12.01**	14.92
MK07	186.62	223.77	420.75	398.25	8.96	**8.72**
MK08	512.57	413.46	507.68	494.03	17.01	**16.99**
MK09	464.19	429.99	523.07	421.88	17.41	**16.98**
MK10	563.42	533.41	579.99	456.16	**18.49**	19.21
DP01a	329.58	378.70	552.35	461.91	**14.37**	15.04
DP02a	399.48	360.97	551.51	463.61	15.30	**15.11**
DP03a	372.3	342.92	567.21	451.94	**15.07**	15.48
DP04a	351.47	397.92	566.65	451.82	15.06	**13.43**
DP05a	363.12	369.17	552.73	468.03	16.18	**16.09**
DP06a	352.44	430.88	561.88	454.19	**15.02**	15.50
DP07a	434.88	441.16	611.62	527.73	17.13	**17.05**
DP08a	424.21	498.77	640.87	528.69	**17.14**	17.78
DP09a	435.24	492.05	612.58	529.86	**17.10**	17.65
DP10a	408.32	482.17	602.94	532.11	16.25	**16.15**
DP11a	416.02	533.27	613.92	531.21	**16.39**	16.51
DP12a	467.75	536.83	629.338	513.90	18.87	**18.58**
DP13a	618.31	691.63	704.50	610.83	**19.48**	19.55
DP14a	629.81	606.47	708.58	613.89	19.63	**19.49**
DP15a	618.89	608.36	718.54	582.27	20.24	**20.19**
DP16a	631.26	708.18	711.19	598.37	**19.55**	20.15
DP17a	682.38	637.38	705.61	589.20	20.34	**20.17**
DP18a	678.05	648.08	730.39	588.57	20.17	**19.81**
better	0	0	0	0	13	**15**

of FSL parameters: $0.5\ m, m$, and $2m$. Combining these parameters, we compare the performance of DSGRL with MOEA/D (baseline), INSBBO (best performing method in comparative trials), and PGRL (a state-of-the-art multi-objective GRL method) on MK10. Because tuning the IBDR parameters directly affects the optimization level of the three objectives, we employ solution set coverage (SC) to visually compare the multi-objective dominance among the resulting solution sets. Specifically, $SC(A, B)$ measures the proportion of solutions in B that are dominated by those in A. The value of SC ranges from 0 to 1, and values closer to 1 indicate higher solution set quality.

Table 6 presents the solution results under various parameter levels. Here, "P1F1" indicates that both the performance period parameter and the FSL parameter are set to the first element in their respective parameter sets, and subsequent symbols follow the same pattern. From the table, it is evident that DSGRL achieves the optimal SC for at least 6 parameter combinations. Moreover, a longitudinal comparison of the

Table 6
SC results for different combinations of IBDR parameters.

No.	DSGRL(G1) vs MOEA/D(M1)		DSGRL(G1) vs INSBBO(I1)		DSGRL(G1) vs PGRL(G2)	
	SC(G1,M1)	SC(M1,G1)	SC(G1,I1)	SC(I1,G1)	SC(G1,G2)	SC(G2,G1)
P1F1	0.11	**0.44**	0.15	**0.46**	**0.39**	0.23
P1F2	**0.38**	0.28	0.04	**0.29**	**0.63**	0.18
P1F3	**0.54**	0.32	**0.34**	0.31	**0.6**	0.15
P2F1	**0.41**	0.35	0.11	**0.25**	**0.38**	0.28
P2F2	**0.48**	0.27	**0.26**	0.18	**0.66**	0.11
P2F3	**0.53**	0.16	**0.43**	0.39	**0.63**	0.12
P3F1	**0.39**	0.23	**0.16**	0.10	**0.45**	0.34
P3F2	**0.51**	0.26	**0.39**	0.11	**0.76**	0.05
P3F3	**0.58**	0.2	**0.41**	0.23	**0.95**	0.01
better	8	1	6	3	9	0

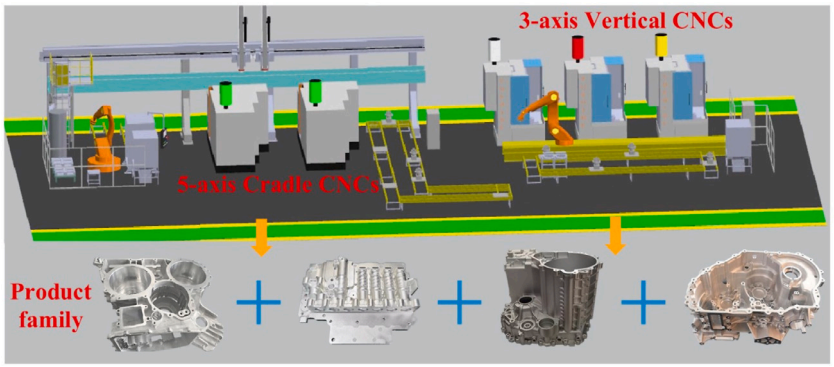


Fig. 8. Five processing machines and four categories of products in a mechanical parts prototype workshop.

two parameters (i.e., "P1F1" vs. "P1F2" and "P1F1" vs. "P2F1") reveals that, as the IBDR policy becomes more lenient, the advantage of our algorithm becomes increasingly pronounced. DSGRL is inferior to both heuristic approaches only when both the performance period parameter and the FSL parameter are set to their smallest values. In this strict load-shedding scenario, fewer scheduling options enable cutting, allowing search-based heuristic algorithms to find certain extreme solutions by sacrificing solution time.

This observation suggests a compelling avenue for future research: by incorporating relevant domain knowledge (e.g., right-shift during the performance period) into the state features, we may guide GRL to learn appropriate strategies for such stringent conditions. Overall, the proposed algorithm effectively balances makespan, TEC, and TPL under various IBDR configurations. These results demonstrate the algorithm's robustness in managing diverse scenarios, thereby providing decision-makers with a wide range of satisfactory solutions.

5.5. Case study

This paper conducts an application validation in a prototype workshop for mechanical components located in Anhui Province, China. The workshop provides customers with high-precision prototyping services for single-piece, small-lot, and multi-variety prototypes. As illustrated in Fig. 8, a specific part of the workshop is selected for numerical analysis, encompassing five CNC machines used to manufacture four major product categories. Experiments are performed using data from ten orders within a production cycle in August 2023. Details concerning the processes and energy consumption are presented in D. Drawing on the experimental methodologies referenced in Füchtenhans and Glock (2024), Li and Hong (2016), we define three scenarios for which solutions are independently derived:

1)Scenario 1 (Without IBDR): In this scenario, IBDR and its associated power constraints are not considered, transforming the original problem into a bi-objective optimization problem that prioritizes makespan and TEC.

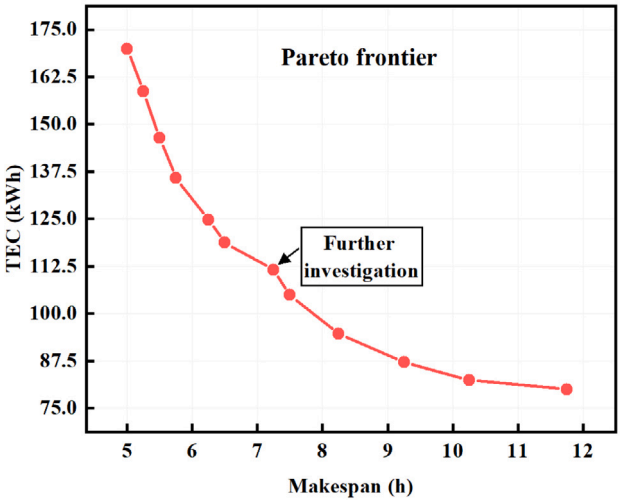
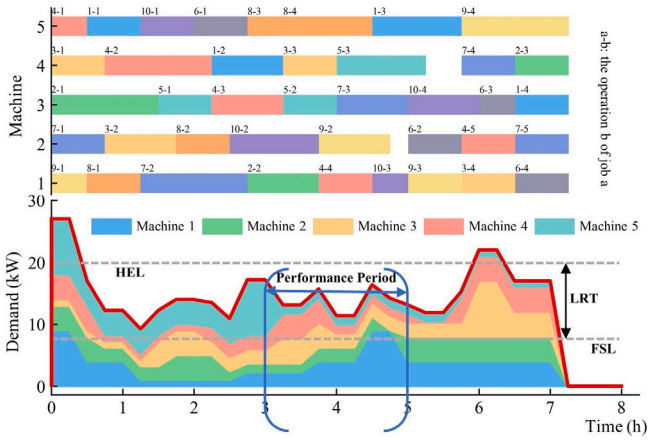


Fig. 9. Approximate Pareto frontier derived from Scenario 1.

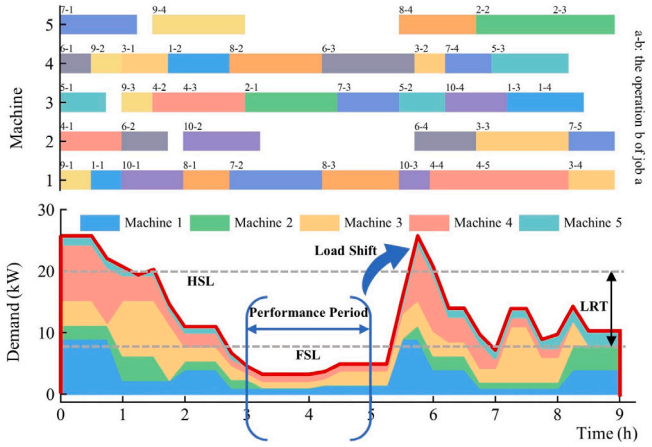
Following (Füchtenhans & Glock, 2024), we utilize a constraint-based approach, designating TEC as a constraint and optimizing makespan using the commercial solver Gurobi 10.0.3, with a solution time limit of 1200 s. By iteratively tightening the constraints, the approximate Pareto frontier shown in Fig. 9 is achieved. From this Pareto frontier, we select a solution that balances the two objectives as a baseline for deeper study into IBDR (noted as "further investigation" in the figure). The optimized values of the scheduling scheme are listed in the second column of Table 7, with the associated Gantt chart and machine-specific energy consumption trajectories illustrated in the upper and lower sections of Fig. 10(a), respectively. The trajectories curve is segmented into four subnodes per unit span of the horizontal axis, corresponding to a time unit of 15 min.

Table 7
Optimization objective values for benchmark cases in Scenarios 1–3.

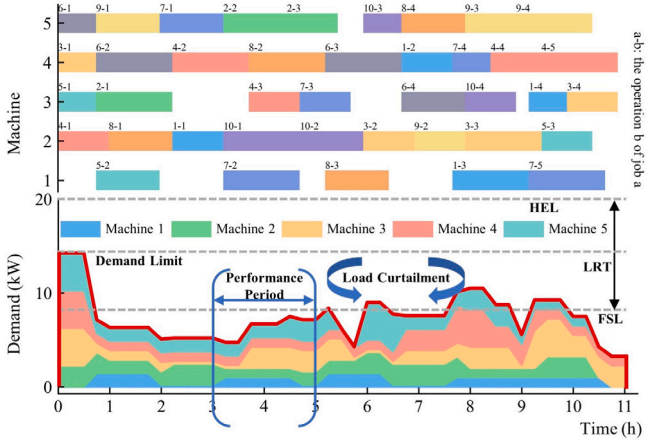
Instances	Scenario1	Scenario2	Scenario3
Makespan (h)	7.25	9	11
TEC (kWh)	111.58	110.42	82.93
TPL (kW)	–	30.76	49.38



(a) Gantt chart and load profiles for the baseline in Scenario 1.



(b) Gantt chart and load profiles for the baseline in Scenario 2.



(c) Gantt chart and load profiles for the baseline in Scenario 3.

Fig. 10. Gantt chart and machine-specific energy consumption trajectories for the baselines in Scenarios 1–3.

2)Scenario 2 (With IBDR & Load Deferral): Now consider the DA-PSI event, which starts at the third hour and lasts for two hours ($L = 8$). Other key parameters include: (1) HEL at 20 kW, (2) LRT at 12 kW, (3) an incentive rate of 3 CNY/kW. Here, electrical demand can be deferred without additional constraints (Füchtenhans & Glock, 2024).

The three-objective Pareto-optimal frontier obtained by the multi-objective GNN schedulers is shown in the top left corner of Fig. 11(a) which demonstrates that the non-dominated scheduling solutions form an extensive yet finite set. Analogous to Scenario 1, the Makespan-TEC view is plotted in the bottom left corner of Fig. 11(a), where the benchmark solution for further comparison is marked. The optimized values for the new baseline are detailed in the third column of Table 7. Compared to Scenario 1, the new scheme extends the makespan by 1.75 h but secures $(20 \cdot 8 - 30.76) \cdot 3 = 387.7$CNY in incentives.

The updated Gantt chart and load profiles are depicted in Fig. 10(b). Unlike the Gantt chart in Scenario 1, this scenario prioritizes allocating machines with longer processing times (corresponds to lower power consumption) or setting machines to standby during the performance period. For instance, during the performance period, operation $O_{8,2}$, $O_{6,3}$, $O_{2,1}$ and $O_{7,2}$, each with a processing time of 6, begin execution, while machines M_2 and M_5 remain in standby mode. This strategy successfully shifts the load during the performance period to the $[5, 6]$ time slot to secure incentives, as evidenced by the peak convex points on the demand curve.

3) Scenario 3 (With IBP & Load curtailment): Building on Scenario 2, this scenario introduces an additional constraint: a maximum power limit throughout the cycle, set at 16 kW, thereby restricting arbitrary deferral of power demand to other periods (Füchtenhans & Glock, 2024).

The newly calculated Pareto-optimal frontier and its associated Makespan-TEC view are depicted in Fig. 11(b). Relative to Scenario 2, the number of non-dominated solutions is sparser, indicating that the introduction of maximum power constraints significantly narrows the feasible solution space. Notably, this scenario results in a longer average makespan for all scheduling schemes but adheres strictly to each IBDR-defined constraint on energy.

As in previous scenarios, reference points are set, and the resulting Gantt chart and load profiles are illustrated in Fig. 10(c). Analysis of Gantt charts reveals that the new scheme reduces peak demand by adjusting machine states and allocations within $[1, 10]$ time slot, thus balancing energy consumption across the production cycle (as evidenced by a relatively flatter power consumption curve). The last column of Table 7 presents the optimized objective values for the benchmark scheme in this scenario. Compared to Scenario 2, to meet the additional power constraint, the makespan is further sacrificed, with a relative decrease in TEC of up to 25%.

Fig. 12 summarizes the peak demand and average load across the three scenarios. It is apparent that, compared to scenarios without IBDR, the remaining scenarios exhibit improved performance on energy metrics, with Scenario 3 achieving the reduction of 47.22% in peak demand and 51.01% in average load. Considering the increasing importance of reducing peak demand for grid stability, it is essential to balance the extension of makespan within reasonable limits to capitalize on the benefits provided by Scenario 3.

6. Conclusion and future work

This paper explores the energy-efficient scheduling problem in flexible job shops under IBDR, introducing a multi-objective GRL approach for the synergistic optimization of production and energy. During the problem modeling phase, an MDP is developed for EFJSP-IBDR, characterized by scheduling states represented through an improved heterogeneous disjunctive graph and four sets of raw features. Additionally, a reward function is constructed using a bilayer aggregation function of trajectory and action. In the problem-solving phase, a novel multi-objective GNN scheduler comprising modules for

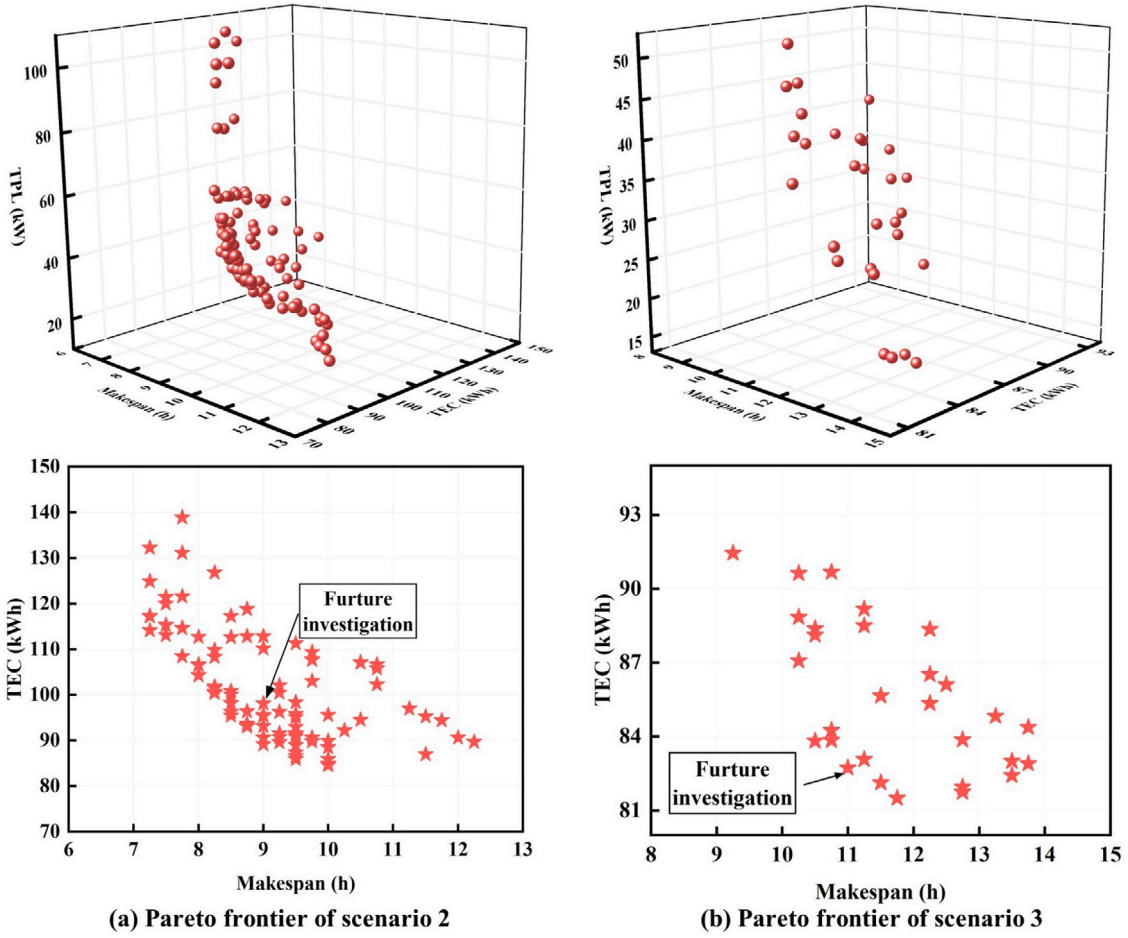


Fig. 11. Approximate Pareto frontier distribution diagrams for Scenarios 2 and 3.

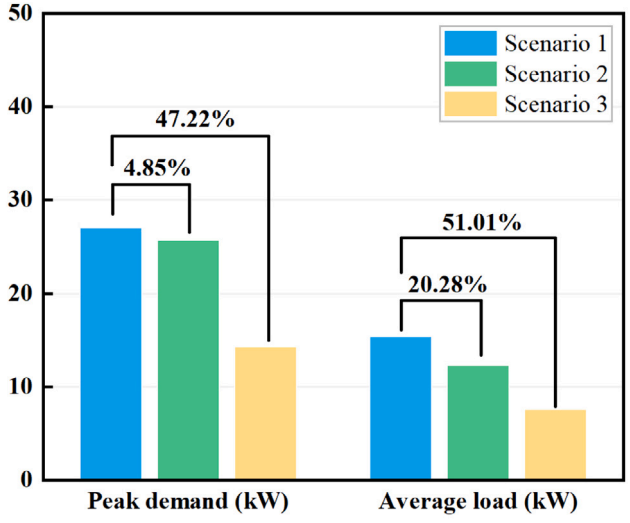


Fig. 12. Load conditions for benchmark cases across three scenarios.

representation learning and policy learning is devised. This scheduler transforms high-dimensional graph features into low-dimensional embeddings via metapath-edge attention and heterogeneous attention mechanisms and employs a similarity-based parameter transfer method to train scheduling strategies under various weight vectors.

The results of the case study in a mechanical parts prototype workshop indicates that the proposed scheduling model can reduce the peak demand by 47.22% and reduce indirect emissions by 51.01%. Moreover, rigorous studies and analyses of standard benchmarks are conducted. The results demonstrate that the proposed method outperforms some well-known algorithms in terms of solution quality and runtime. Given the escalating pressures of energy consumption and the growing mismatch between power supply and demand, industrial users are expected to increasingly focus on optimizing power consumption patterns. This paper provides accurate insights for workshop managers to analyze load-side flexibility and make cooperative production and energy decisions. Additionally, our method also delivers essential data on load reduction potential to utility operators for designing flexible incentive policies tailored to different industrial consumers.

Several issues merit further research. First, in the context of multi-variety small-batch production, the time and energy consumption for switching between products demand special attention. Incorporating these factors into the modeling process could enhance the universality and practical applicability of energy-efficient scheduling methods. Furthermore, as green energy sources such as photovoltaics and wind, along with energy storage technologies, become increasingly prevalent, their impact on load-side production and energy collaborative optimization is substantial and warrants careful consideration. Finally, yet importantly, the algorithm described in this paper necessitates the training of numerous GRL models for various subproblems. Despite the implementation of the acceleration strategy, these models still impose a substantial computational load. Future research could explore leveraging the generalization capabilities of deep learning models to address related subproblems with a single model, potentially simplifying the approach to approximating the Pareto frontier.

Table A.8
Notation and definitions.

Notation	Definition
Indices	
n	The number of jobs
i	The index of a job, $i \in J = 1, 2, \ldots, n$
m	The number of machines
k	The index of a machine, $k = 1, 2, \ldots, m$
j	The index of an operation in job i, $j = 1, 2, \ldots, n_i$
u	The number of time periods
l	The index of a time period, $l = 1, 2, \ldots, u$
Parameters	
n_i	The number of operations for job i
O_{ij}	The jth operation of job i
M_{ij}	The set of eligible machines of O_{ij}
m_{ij}	The number of machines in set M_{ij}
t_{ijk}	The processing time of O_{ij} on machine k (time periods)
p_{ijk}	The energy consumption rate of O_{ij} on machine k (kW)
$[ps, pe]$	Performance periods specified by IBDR event, $ps, pe \leq u$
Variables	
S_{ij}	The start time of processing O_{ij} (time periods)
C_i	The completion time of job i (time periods)
Decision Variables	
X_{ijk}	1 if O_{ij} is processed on machine k; otherwise, 0
Y_{ijkl}	1 if O_{ij} is processed on machine k in the lth time period; otherwise, 0
$Z_{iji'j'}$	1 if O_{ij} is processed before $O_{i'j'}$ on the same machine; otherwise, 0
Objectives:	
C_{max}	The maximum completion time (time period)
TEC	Total energy consumption (kWh)
TPL	Total payload (kW)

CRediT authorship contribution statement

Mingzhou Liu: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Investigation. **Zhangjie Rui:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Investigation. **Xi Zhang:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing. **Maogen Ge:** Resources, Supervision, Project administration, Data curation. **Lin Ling:** Supervision, Funding acquisition. **Xiaoqiao Wang:** Writing – review & editing, Funding acquisition. **Conghu Liu:** Project administration.

Funding

This work was supported by the National Natural Science Foundation of China [grant number 72401085], the Fundamental Research Funds for the Central Universities of China [grant number PA2024GDSK0041] and the Projects of International Cooperation and Exchanges NSFC, China [grant number W2411063].

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lin Ling reports financial support was provided by Fundamental Research Funds for the Central Universities of China. Xiaoqiao Wang reports financial support was provided by National Natural Science Foundation of China. Mingzhou Liu reports financial support was provided by Projects of International Cooperation and Exchanges NSFC. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Notation and definitions in problem formulation

See Table A.8.

Appendix B. Details of the raw feature design

Raw features of operation node: For each O_{ij}, the raw feature vector μ_{ij} has seven elements:

(1) status: 1 if O_{ij} is scheduled, otherwise 0;
(2) number of neighboring nodes: $\left|\mathcal{N}_t\left(O_{ij}\right)\right|$;
(3) number of unscheduled operations in the job: the number of operations in J_i that have not been scheduled;
(4) start time: the start time of O_{ij} at the current time step t;
(5) completion time: the completion time of O_{ij} at the current time step t;
(6) remaining workload: the total processing time for unscheduled operations in J_i;
(7) processing energy consumption: the processing energy consumption of O_{ij} at the current time step t.

Raw features of machine node: For each M_k, four elements are designed to form its raw feature vector υ_k:

(1) number of neighboring nodes: total number of nodes in graph topology neighbors and metapath neighbors;
(2) available time: the time when M_k completes all its assigned operations till time step t;
(3) standby time: cumulative standby time of M_k till time step t;
(4) standby energy consumption: cumulative standby energy consumption of M_k till time step t.

Raw features of disjunctive arc: The processing time t_{ijk} and power p_{ijk} corresponding to the disjunctive arc together form its raw feature vector δ_{ijk}.

IBDR features: Three elements are used to build IBDR features ψ:

(1) number of operations: total count of operations that intersect between their processing time window and the performance period;

Table C.9
Taguchi experiments for DSGRL.

No.	GNN layer	Hidden dim	Learning rate	Discount factor	RV
1	1	64	2.5×10^{-2}	0.94	−54.47
2	1	128	2.5×10^{-3}	0.96	−42.09
3	1	256	2.5×10^{-4}	0.98	−26.30
4	2	64	2.5×10^{-3}	0.98	−56.93
5	2	128	2.5×10^{-4}	0.94	−37.27
6	2	256	2.5×10^{-2}	0.96	−28.11
7	3	64	2.5×10^{-4}	0.96	−26.84
8	3	128	2.5×10^{-2}	0.98	−104.24
9	3	256	2.5×10^{-3}	0.94	−82.49

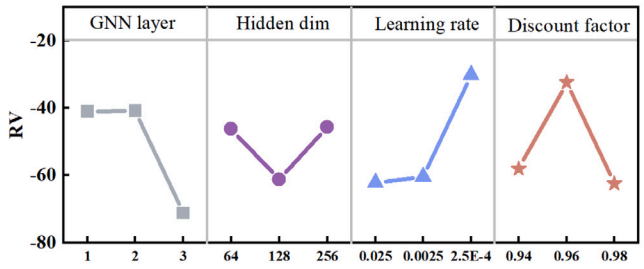


Fig. C.13. Parameter level trends chart.

(2) number of successful incentives: count of time windows that achieve the load reduction conditions;
(3) average load: average power across all time windows that fulfill the load reduction criteria.

Appendix C. The sensitivity analysis of several key hyperparameters

Since our scheduling model includes numerous adjustable hyperparameters, we preset the PPO hyperparameters to values that perform well in classic RL tasks to limit the number of sensitivity experiments. Only the hyperparameters related to network training are adjustable in our experiments. First, the value ranges of the hyperparameters are as follows: the number of iterations for the representation learning network $[1, 2, 3]$, the hidden dimensions of the MLP $[64, 128, 256]$, the learning rate $[2.5\times10^{-2}, 2.5\times10^{-3}, 2.5\times10^{-4}]$ and the discount factor $[0.94, 0.96, 0.98]$. Then all combinations of four parameters could be obtained and the orthogonal array $L_9(3^4)$ is constructed. Taguchi method is utilized to perform the parameter experiments and the average cumulative reward obtained from all parallel instances across different training time steps is considered as the response value (RV) of the corresponding parameter combination. All models are trained for $0.5-1$ hours, and the FJSP environment parameters are set to $[n=10, m=5, u=96]$.

The obtained RV values for different parameter combinations based on the Taguchi method are presented in Table C.9. From Table C.9, it is evident that groups with larger learning rate parameters yield significantly worse results compared to other groups, indicating that the algorithm's performance is highly sensitive to the learning rate. By calculating the average RV at various parameter levels, we generated a plot of parameter level trends, as shown in Fig. C.13. Evidently, larger RV values correspond to better parameter combinations. Thus, the selected parameters for the scheduling model are $[2, 256, 2.5\times10^{-4}, 0.96]$.

Appendix D. Production and energy data from the real case

See Table D.10

Table D.10
The processing times and power consumption data for ten orders in the actual case study.

J_i	O_{ij}	t_{ijk} (15-min)/ p_{ijk} (kW)				
		$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
J_1	O_{11}	2/9	4/2.25	–	–	3/4
	O_{12}	–	1/36	–	4/2.25	–
	O_{13}	6/1	–	2/9	–	5/1.44
	O_{14}	–	–	3/4	–	2/9
J_2	O_{21}	–	3/4	6/1	–	–
	O_{22}	4/2.25	–	1/36	–	5/1.44
	O_{23}	3/4	2/9	–	3/4	4/2.25
J_3	O_{31}	1/36	–	–	3/4	2/9
	O_{32}	–	4/2.25	–	2/9	–
	O_{33}	–	6/1	2/9	3/4	5/1.44
	O_{34}	3/4	2/9	4/2.25	–	2/9
J_4	O_{41}	–	4/2.25	–	–	2/9
	O_{42}	3/4	–	2/9	6/1	–
	O_{43}	–	–	4/2.25	–	–
	O_{44}	3/4	–	–	4/2.25	1/36
	O_{45}	6/1	3/4	5/1.44	6/1	3/4
J_5	O_{51}	–	5/1.44	3/4	–	–
	O_{52}	5/1.44	–	3/4	–	6/1
	O_{53}	2/9	4/2.25	–	5/1.44	4/2.25
J_6	O_{61}	1/36	–	–	2/9	3/4
	O_{62}	–	3/4	–	6/1	–
	O_{63}	–	5/1.44	2/9	6/1	1/36
	O_{64}	3/4	4/2.25	5/1.44	–	1/36
J_7	O_{71}	–	3/4	–	–	5/1.44
	O_{72}	6/1	–	3/4	2/9	–
	O_{73}	–	–	4/2.25	–	–
	O_{74}	1/36	–	–	3/4	2/9
	O_{75}	6/1	3/4	5/1.44	6/1	3/4
J_8	O_{81}	3/4	5/1.44	–	–	4/2.25
	O_{82}	–	3/4	–	6/1	–
	O_{83}	5/1.44	–	1/36	–	2/9
	O_{84}	–	–	3/4	–	5/1.44
J_9	O_{91}	2/9	1/36	–	–	5/1.44
	O_{92}	–	4/2.25	–	2/9	–
	O_{93}	3/4	–	2/9	–	4/2.25
	O_{94}	–	–	1/36	–	6/1
J_{10}	$O_{10,1}$	4/2.25	6/1	–	–	3/4
	$O_{10,2}$	–	5/1.44	–	2/9	–
	$O_{10,3}$	2/9	–	2/9	–	3/4
	$O_{10,4}$	–	–	4/2.25	–	3/4

Data availability

We have published the code in the experimental section.

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